

Insecure protocols	Secure protocols	Transport layer	Use cases
TCP or UDP	SSL/TLS	Transport layer	
IP	IPSec - AH/ESP and Tunnel/ Transport	Network layer	
FTP	SFTP / FTPS	TCP	File transfer protocol
Telnet	SSH	TCP	Remote access
DNS	DNSSEC	TCP/UDP	Domain name resolution
HTTP	HTTPS/SHTTP	TCP	Web
NTP	NTS	TCP	Time synchronisation
SNMP	SNMPv3	TCP	Network monitoring
RTP	RTPS	UDP (mostly)	Voice and video
LDAP	LDAPS	TCP	Directory services
MIME, POP, IMAP	S/MIME, POP3S, IMAPS	TCP	Email subscription services

- **SMTP or Simple Mail Transfer Protocol** was developed to send emails over port 25.
- **MIME or Multipurpose Internet Mail Extension Protocol** was introduced to allow transfer of non-textual data such as videos, images, audio, and so on.
- **S/MIME or Secure/Multipurpose Internet Mail Extension Protocol** was developed to enhance communication security, using underlying SSL principles.
- **POP or Post Office Protocol and IMAP or Internet Message Access Protocol** were developed for receiving email. Here the login credentials are transferred over a plain text network for incoming emails.
- **POP3S - port 995 and IMAPS - port 993** were implemented allowing SSL/TLS in the transport layer.

LDAP vs LDAPS

LDAP or Lightweight Directory Access Protocol is a directory service used for fetching user and host information quickly. By default, LDAP is not secure, leading to the development of a secure version called LDAPS.

LDAPS uses SSL/TLS in the transport layer. This is how it works. The LDAP session starts between a client and the LDAP server, commonly referred to as the Directory System Agent (DSA), on default port 636. The server sends a client operation request response — there are various requests a client can make such as bind, search, and entry editing.

The DSA or LDAP server is the database that stores information hierarchically for easy retrieval.

RTP vs SRTP

The RTP protocol is used to transfer audio and video communications. As it transfers data in raw format, it is vulnerable. The SRTP protocol was developed as an extension to provide security by implementing SSL/TLS in the transport layer.

SNMP vs SNMPv3

SNMP or Simple Network Management Protocol is used to collect statistics of devices in the network. This protocol helps

to monitor the health of systems, but is not secure.

SNMP version 3 or SNMPv3 was built to overcome these challenges. It was preceded by SNMPv1 and SNMPv2 but these were built primarily for performance and not security.

NTP vs NTS

NTP or Network Time Protocol is used to synchronise time within the devices with the network time server. It has the same weakness as other non-secure protocols. To overcome this, NTS has been developed with a similar mechanism by including SSL/TLS in the transport layer.

The table above summarises the secure and insecure protocols we have briefly touched upon so far.

We have now learnt about the secure transfer of data and how it is sent to the intended recipient. As a key takeaway, it is essential to use secure protocols at every possible layer to ensure maximum security. Stay safe and secure! 

References

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The author is a lifelong learner influenced by the quote:
"The more I learn, the more I realise how much I don't know."
 -Albert Einstein